

**Department of Computer Science and Engineering**

**B.Tech III Semester**

**Academic Year 2026-2027**

**Database Management Systems(2501IT05)**

**Capstone Project Documentation**



**ADITYA UNIVERSITY**

**Surampalem, Kakinada District**

**Andhra Pradesh-533437**



# ADITYA UNIVERSITY

**SCHOOL OF COMPUTING**

**Department of Computer Science and Engineering**

## **CERTIFICATE**

This is to certify that the Cornerstone project work entitled “**Employee Payroll Management System**” is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2026-2027.

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## DECLARATION

We hereby declare that the Cornerstone project entitled “**Employee Payroll Management System Application**” is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. Degree in Computer Science and Engineering**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

Place : Surampalem

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## ACKNOWLEDGEMENT

It is with immense pleasure that we express our sincere gratitude to our Cornerstone Project Guide, **Mrs. K Vydehi, Assistant Professor**, who has guided us throughout the project. Her valuable support, encouragement, and guidance have greatly contributed to the successful completion of this work.

We would like to thank our Deputy Heads of the Department, **Dr. Chandra Sekhar Kolli, Associate Professor, Dr Hari Jyothula, Associate Professor and Dr Pennada Siva Satya Prasad, Assistant Professor**, Department of Computer Science and Engineering, for their support and valuable suggestions.

Our deepest thanks to our Head of the Department **Dr. Phani Sridhar Addepalli, Assistant Professor**, for his inspiration and for providing all the necessary facilities and resources required for this project.

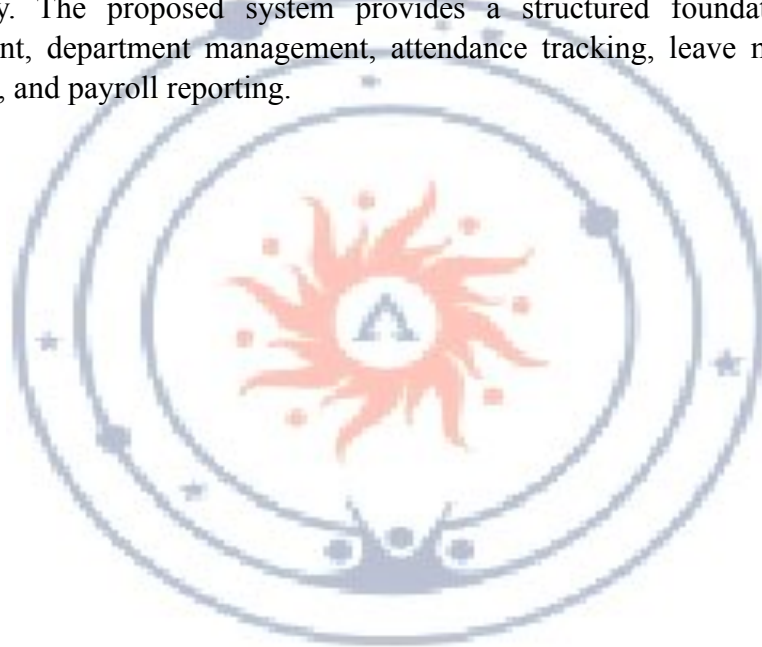
We also extend our sincere thanks to **Dr. M.V. Rajesh, Associate Dean, School of Computing** for their encouragement and support during the course of our project.

We would like to express our sincere gratitude to **Dr. G. Suresh, Registrar, Dr. A Ramesh, Pro Vice-Chancellor (Engineering & Sciences), Dr. S.RamaSree, Pro Vice-Chancellor (Academics), Dr. M.B. Srinivas, Vice-Chancellor, Dr. M. Sreenivasa Reddy, Deputy Pro Chancellor and Management, Aditya University** for providing excellent infrastructure and state-of-the-art laboratories.

Finally, we would like to thank the **faculty members, lab technicians, non-teaching staff, and our friends** who have directly or indirectly supported us in completing this Cornerstone project successfully.

## ABSTRACT

The Employee Payroll Management System is designed to automate and manage employee payroll activities within an organization through a centralized relational database. The system stores and manages information related to departments, employees, attendance, leaves, and payroll details. The database minimizes redundancy, maintains data integrity through primary and foreign keys, and supports efficient retrieval of employee and payroll information using SQL queries. The design is normalized up to Third Normal Form (3NF), helping reduce update anomalies and improve data consistency. The proposed system provides a structured foundation for employee management, department management, attendance tracking, leave management, salary processing, and payroll reporting.



## LIST OF FIGURES

Figure 1. ER Diagram of Employee Payroll Management Database



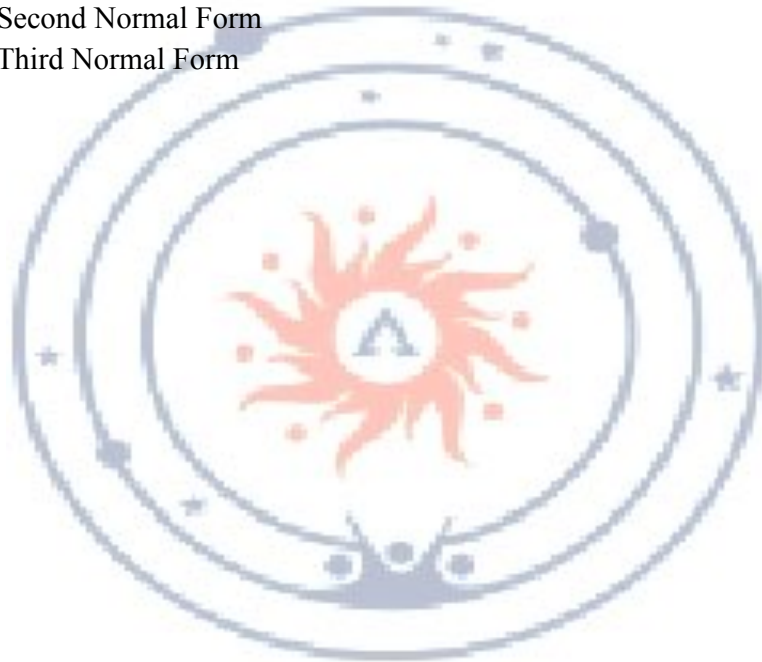
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## ACRONYMS

- DBMS – Database Management System
- ER – Entity Relationship
- SQL – Structured Query Language
- DDL – Data Definition Language
- DML – Data Manipulation Language
- DQL – Data Query Language
- DCL – Data Control Language
- TCL – Transaction Control Language
- PK – Primary Key
- FK – Foreign Key
- 1NF – First Normal Form
- 2NF – Second Normal Form
- 3NF – Third Normal Form



# 1. INTRODUCTION

## 1.1 Brief Information about the Project

An Employee Payroll Management System is a database application that manages employee information and payroll-related activities within an organization. The project provides a centralized relational database for departments, employees, attendance, leave, payroll, and salary-related information. It is intended to replace fragmented manual records and multiple spreadsheet files with a structured database that supports consistent data storage, accurate payroll processing, and efficient retrieval of employee and salary information.

## 1.2 Motivation and Contribution of Project

The project is motivated by common problems in employee and payroll management such as duplicate records, manual data entry, time-consuming searches, calculation errors, and difficulty in maintaining accurate payroll information. A centralized database provides a systematic approach to storing employee-related data and maintaining relationships between departments, employees, attendance, leave, and payroll records through keys and constraints.

- Centralizing employee and payroll records in one relational database.
- Reducing duplicate and inconsistent employee information.
- Maintaining referential integrity using primary and foreign keys.
- Supporting accurate and faster payroll-related queries and report generation.
- Providing a structured foundation for department, employee, attendance, leave, and payroll management.

## 1.3 Objectives of the Project

- Develop a centralized employee payroll database.
- Manage employee records efficiently.
- Maintain department details.
- Store and manage employee attendance information.
- Manage employee leave records.
- Maintain employee salary and payroll details.
- Reduce duplicate data and update anomalies.
- Maintain data integrity using primary and foreign keys.
- Support efficient SQL queries and payroll report generation.

## 1.4 Scope of the Project

The scope covers database-oriented management of employee and payroll information within an organization. The core scope includes department, employee, attendance, leave,

payroll, and salary-related data. The database can be used as the backend for an administrative application and can be extended later with employee performance, benefits, tax management, overtime tracking, loan management.



## 2. SYSTEM ANALYSIS

### 2.1 Existing System

- Manual maintenance of employee and payroll records.
- Use of multiple Excel sheets or disconnected files for storing information.
- Duplicate employee records and inconsistent payroll information.
- Time-consuming searches for employee, attendance, leave, and salary details.
- Difficulty in generating consolidated payroll and employee reports.
- Higher possibility of data-entry and salary calculation errors.
- Difficulty in maintaining relationships between employee, department, attendance, leave, and payroll records.

### 2.2 Proposed System

The proposed system uses a centralized relational database to manage employee and payroll-related information. It organizes employee data into related tables and uses primary keys, foreign keys, uniqueness rules, NOT NULL conditions, and CHECK constraints where appropriate. The system supports department, employee, attendance, leave, payroll, and salary management while enabling efficient SQL-based data retrieval and reporting.

- Centralized employee payroll database
- Department management
- Employee management
- Attendance management
- Leave management
- Payroll and salary processing
- SQL-based reporting
- Data integrity and referential integrity

## 2.3 Functional Requirements

**Table-1 : Functional Requirements**

| Module                | Functional Requirement                                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Department Management | Store, update, search and maintain department information.                                                                 |
| Employee Management   | Add, update, search, and display Employee details.                                                                         |
| Attendance Management | Record and maintain employee attendance, check-in, check-out, and attendance status.                                       |
| Leave Management      | Store and manage employee leave applications, leave types, dates, reasons, and leave status.                               |
| Job Management        | Store and maintain job information such as Job ID, Job Title, Basic Salary, and Job Description.                           |
| Payroll Management    | Maintain salary details including basic salary, bonus, deductions, and net salary.                                         |
| Search and Reporting  | Retrieve employee, attendance, leave, department, and payroll information using SQL queries and generate relevant reports. |
| Data management       | Maintain data integrity and consistency using primary keys, foreign keys, and other database constraints.                  |

## 2.4 Non-Functional Requirements

- Accuracy – Payroll information, salary components, attendance, leave, and employee records should be maintained correctly.
- Security – Employee and salary information should be protected from unauthorized access.
- Reliability – The system should work consistently and maintain reliable employee and payroll records.
- Performance – Employee, attendance, leave, and payroll information should be retrieved efficiently.
- Usability – The system should be simple and understandable for HR/administrators, payroll/accounts staff, and management.
- Data Integrity – Employee, department, attendance, leave, and payroll records should remain consistent and accurate.
- Maintainability – The database should be easy to maintain and update as employee and payroll information changes.

## 2.5 Requirements Specification

**Table-2 : Requirement Specification**

| Category         | Suggested Requirement                                                                       |
|------------------|---------------------------------------------------------------------------------------------|
| Hardware         | Intel i3 or above; 4 GB RAM minimum; adequate storage; standard display and keyboard/mouse. |
| Operating System | Windows 10/11 or an equivalent supported platform.                                          |
| Database         | Oracle Database or another relational DBMS used by the implementation.                      |



## 3. DATABASE DESIGN

### 3.1 Major Entities

**Table-3 : Major Entities**

| Entity     | Description                            |
|------------|----------------------------------------|
| Department | Stores department information.         |
| Employee   | Stores employee records.               |
| Attendance | Stores employee attendance records.    |
| Leave      | Stores employee leave information.     |
| Payroll    | Maintains employee salary information. |
| Job        | Stores job-related information         |

### 3.2 ER Model

The ER model represents the major entities and their relationships in the Employee Payroll Management System. The ER diagram includes Department, Employee, Attendance, Leave, Payroll, and Job entities. The main relationships shown are Department–Employee, Employee–Attendance, Employee–Leave, Employee–Payroll, and Employee–Job. The diagram also includes attributes such as Department\_ID, Department\_Name, Location, Employee\_ID, Employee\_Name, Phone, Email, Attendance\_ID, Attendance\_Date, CheckIn, CheckOut, Status, Leave\_ID, LeaveType, LeaveStatus, Reason, StartDate, EndDate, Payroll\_ID, SalaryMonth, BasicSalary, Bonus, Deduction, NetSalary, Job\_ID, JobTitle, and JobDescription.

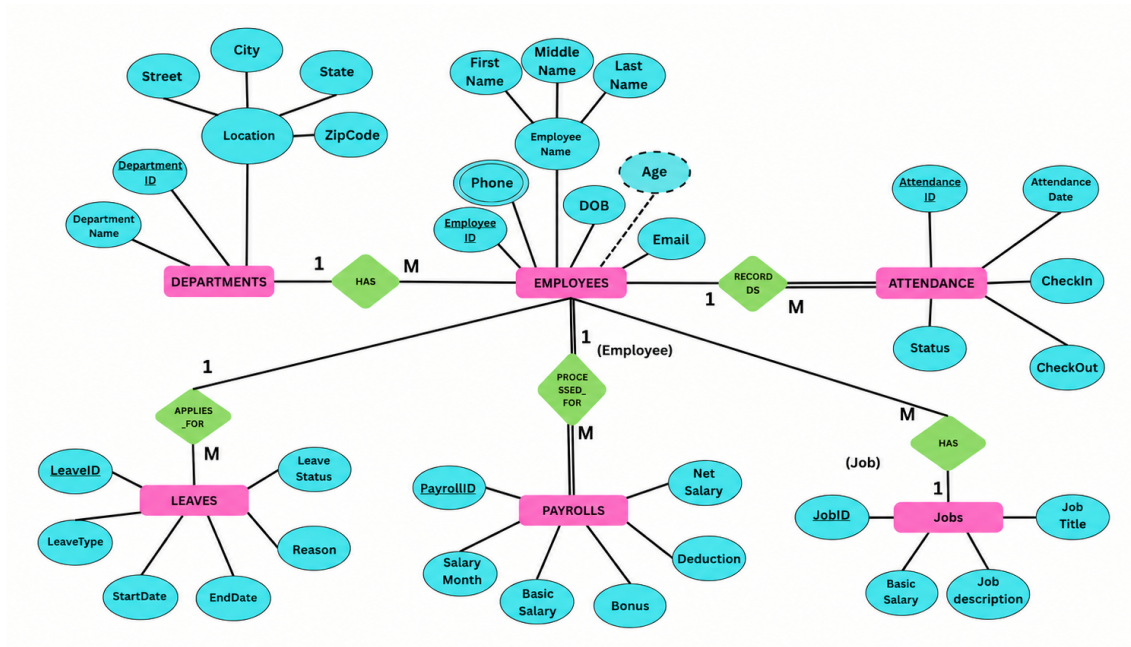
### 3.3 Relationships

**Table-4 : Relationships**

| Relationship          | Description                                                                |
|-----------------------|----------------------------------------------------------------------------|
| Department → Employee | One department can have many employees.                                    |
| Employee → Attendance | One employee can have multiple attendance records.                         |
| Employee → Leave      | One employee can apply for multiple leave records.                         |
| Employee → Payroll    | One employee can have multiple payroll records for different salary months |
| Job → Employee        | One job can be associated with multiple employees.                         |



## 4. ER DIAGRAM



**Figure 1: ER Diagram of Employee Payroll Management Database**

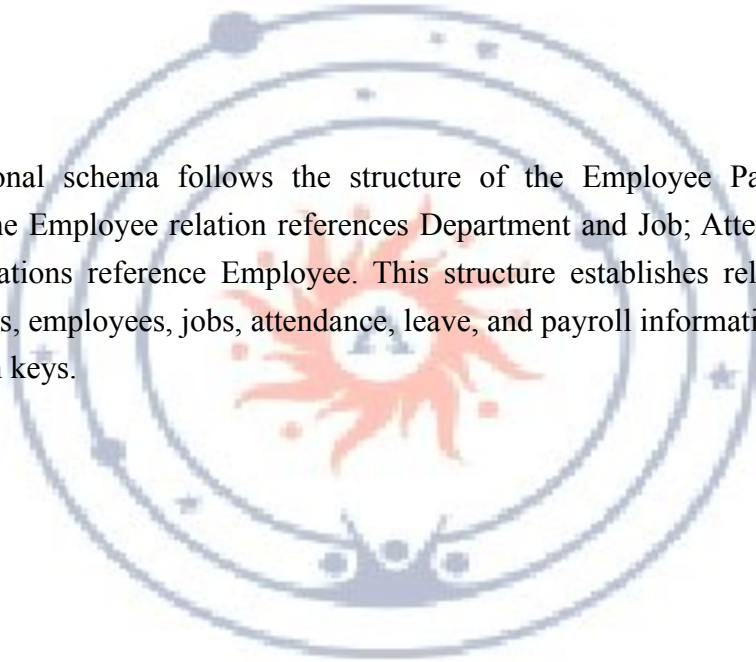
The ER diagram summarizes the entity structure and cardinalities of the Employee Payroll Management System. It represents the relationships among Department, Employee, Attendance, Leave, Payroll, and Job entities based on the proposed database design.

## 5. RELATIONAL SCHEMA

**Table-5 : Relational Schema**

| Relation   | Relational Schema                                                                        |
|------------|------------------------------------------------------------------------------------------|
| DEPARTMENT | Department_ID (PK), Department_Name, Location                                            |
| EMPLOYEE   | Employee_ID (PK), Employee_Name, Phone, Email                                            |
| ATTENDANCE | Attendance_ID (PK), Attendance_Date, CheckIn, CheckOut, Status, Employee_ID (FK)         |
| LEAVE      | Leave_ID (PK), LeaveType, LeaveStatus, Reason, StartDate, EndDate, Employee_ID (FK)      |
| PAYROLL    | Payroll_ID (PK), SalaryMonth, BasicSalary, Bonus, Deduction, NetSalary, Employee_ID (FK) |
| JOB        | Job_ID (PK), JobTitle, BasicSalary, JobDescription, Employee_ID (FK)                     |

The relational schema follows the structure of the Employee Payroll Management System. The Employee relation references Department and Job; Attendance, Leave, and Payroll relations reference Employee. This structure establishes relationships between departments, employees, jobs, attendance, leave, and payroll information through primary and foreign keys.



## 6. TABLE DESIGN

### 6.1 Department Table

Table-6 : Department Table

| Attribute       | Data Type    | Constraint   | Description                  |
|-----------------|--------------|--------------|------------------------------|
| Department_ID   | INT / NUMBER | PK, NOT NULL | Unique department identifier |
| Department_Name | VARCHAR      | NOT NULL     | Department name              |
| Location        | VARCHAR      | —            | Department location          |

### 6.2 Employee Table

Table-7 : Employee Table

| Attribute     | Data Type | Constraint   | Description                |
|---------------|-----------|--------------|----------------------------|
| Employee_ID   | INT       | PK, NOT NULL | Unique employee identifier |
| Employee_Name | VARCHAR   | NOT NULL     | Employee name              |
| Phone         | VARCHAR   | —            | Employee contact number    |
| Email         | VARCHAR   | UNIQUE       | Employee email address     |
| Job_ID        | INT       | FK           | References Job             |
| Department_ID | INT       | FK           | References Department      |

### 6.3 Attendance Table

Table-8 : Attendance Table

| Attribute       | Data Type      | Constraint   | Description                  |
|-----------------|----------------|--------------|------------------------------|
| Attendance_ID   | INT            | PK, NOT NULL | Unique Attendance identifier |
| Attendance_Date | DATE           | NOT NULL     | Date of attendance           |
| CheckIn         | TIME / VARCHAR | —            | Employee check-in time       |
| CheckOut        | TIME / VARCHAR | —            | Employee check-out time      |
| Status          | VARCHAR        | —            | Attendance status            |
| Employee_ID     | INT            | FK           | References Employee          |

## 6.4 Leave Table

**Table-9 : Leave Table**

| Attribute   | Data Type | Constraint   | Description             |
|-------------|-----------|--------------|-------------------------|
| Leave_ID    | INT       | PK, NOT NULL | Unique leave identifier |
| LeaveType   | VARCHAR   | NOT NULL     | Type of leave           |
| LeaveStatus | VARCHAR   | —            | Status of leave request |
| Reason      | VARCHAR   | —            | Reason for leave        |
| StartDate   | DATE      | —            | Leave start date        |
| EndDate     | DATE      | —            | Leave End date          |
| Employee_ID | INT       | FK           | References Employee     |

## 6.5 Payroll Table

**Table-10 : Payroll Table**

| Attribute   | Data Type    | Constraint   | Description                     |
|-------------|--------------|--------------|---------------------------------|
| Payroll_ID  | INT          | PK, NOT NULL | Unique payroll identifier       |
| SalaryMonth | DATE         | NOT NULL     | Salary month                    |
| BasicSalary | INT          | NOT NULL     | Basic salary amount             |
| Bonus       | INT          | —            | Bonus amount                    |
| Deduction   | INT          | —            | Deduction amount                |
| NetSalary   | INT          | —            | Find salary after<br>deductions |
| Employee_ID | INT / NUMBER | FK           | References Employee             |

## 6.6 Job Table

**Table-11 : Job Table**

| Attribute      | Data Type | Constraint   | Description                          |
|----------------|-----------|--------------|--------------------------------------|
| Job_ID         | INT       | PK, NOT NULL | Unique job identifier                |
| JobTitle       | VARCHAR   | NOT NULL     | Job title                            |
| BasicSalary    | INT       | —            | Basic salary associated with the job |
| JobDescription | VARCHAR   | —            | Description of the job               |

## 6.7 Keys and Constraints

**Table-12 : Keys and Constraints Table**

| Constraint Type | Applied To / Purpose                                                                                    |
|-----------------|---------------------------------------------------------------------------------------------------------|
| Primary Key     | Department_ID, Employee_ID, Attendance_ID, Leave_ID, Payroll_ID, Job_ID                                 |
| Foreign Key     | Employee.Department_ID, Employee.Job_ID, Attendance.Employee_ID, Leave.Employee_ID, Payroll.Employee_ID |
| NOT NULL        | Required attributes such as employee names, department names, job titles, and identifiers               |
| UNIQUE          | Employee email and other unique attributes where applicable                                             |
| CHECK           | Values such as salary, bonus, deduction, and attendance status can be restricted to valid values        |
| DEFAULT         | Can be used for fields such as attendance status or payroll components when appropriate                 |

## 7. NORMALIZATION

Normalization organizes data into related relations to reduce redundancy and update anomalies. The supplied project template explicitly requires documentation from UNF to 3NF. The following example uses an academic registration record to illustrate the process.

### 7.1 Unnormalized Form (UNF)

A single employee payroll record might contain Employee\_ID, Employee\_Name, Department\_Name, Job\_Title, Attendance\_Date, Leave\_Type, Salary\_Month, Basic\_Salary, Bonus, Deduction, Net\_Salary, along with multiple attendance, leave, and payroll values in one record. Repeating attendance, leave, and payroll groups and repeated department and job information create data redundancy.

| EMPLOYEE_ID | PAYROLL_MONTH | EMPLOYEE_NAME | DEPARTMENT_ID | DEPARTMENT_NAME | SALARY | BONUS |
|-------------|---------------|---------------|---------------|-----------------|--------|-------|
| 101         | August        | Ravi          | 10            | IT              | 50000  | 5000  |
| 101         | September     | Ravi          | 10            | IT              | 50000  | 6000  |
| 102         | August        | Priya         | 20            | HR              | 45000  | 4000  |
| 102         | September     | Priya         | 20            | HR              | 45000  | 4500  |
| 103         | August        | Arjun         | 10            | IT              | 55000  | 5000  |
| 103         | September     | Arjun         | 10            | IT              | 55000  | 5500  |

### 7.2 First Normal Form (1NF)

In 1NF, repeating groups are removed and each attribute contains atomic values. Each attendance, leave, and payroll record is represented as a separate row. Employee, department, and job information may still be repeated across multiple records.

| EMPLOYEE_ID | EMPLOYEE_NAME | DEPARTMENT_ID | DEPARTMENT_NAME |
|-------------|---------------|---------------|-----------------|
| 101         | Ravi          | 10            | IT              |
| 102         | Priya         | 20            | HR              |
| 103         | Arjun         | 10            | IT              |

| EMPLOYEE_ID | PAYROLL_MONTH | SALARY | BONUS |
|-------------|---------------|--------|-------|
| 101         | August        | 50000  | 5000  |
| 101         | September     | 50000  | 6000  |
| 102         | August        | 45000  | 4000  |
| 102         | September     | 45000  | 4500  |
| 103         | August        | 55000  | 5000  |
| 103         | September     | 55000  | 5500  |

### 7.3 Second Normal Form (2NF)

In 2NF, partial dependencies on a composite key are removed. Employee-specific attributes depend on Employee\_ID, department-specific attributes depend on Department\_ID, job-specific attributes depend on Job\_ID, and attendance, leave, and payroll attributes depend on their respective identifiers. These dependencies motivate separate Department, Employee, Job, Attendance, Leave, and Payroll relations

| EMPLOYEE_ID | EMPLOYEE_NAME | DEPARTMENT_ID |
|-------------|---------------|---------------|
| 101         | Ravi          | 10            |
| 102         | Priya         | 20            |
| 103         | Arjun         | 10            |

| EMPLOYEE_ID | PAYROLL_MONTH | SALARY | BONUS |
|-------------|---------------|--------|-------|
| 101         | August        | 50000  | 5000  |
| 101         | September     | 50000  | 6000  |
| 102         | August        | 45000  | 4000  |
| 102         | September     | 45000  | 4500  |
| 103         | August        | 55000  | 5000  |
| 103         | September     | 55000  | 5500  |

### 7.4 Third Normal Form (3NF)

In 3NF, transitive dependencies are removed. Department and job details are stored separately, while attendance, leave, and payroll records are linked to Employee through Employee\_ID. This reduces redundancy and maintain data consistency.

| Relation   | Reason for Separation                                                                |
|------------|--------------------------------------------------------------------------------------|
| Department | Stores department-specific attributes once.                                          |
| Employee   | Stores employee-specific attributes and references Department and Job.               |
| Job        | Stores job-specific attributes such as job title, basic salary, and job description. |
| Attendance | Stores employee attendance records separately.                                       |
| Leave      | Stores employee leave information separately.                                        |
| Payroll    | Stores salary, bonus, deduction, and net salary information separately.              |

## 8. SQL IMPLEMENTATION

### 8.1 DDL – Data Definition Language

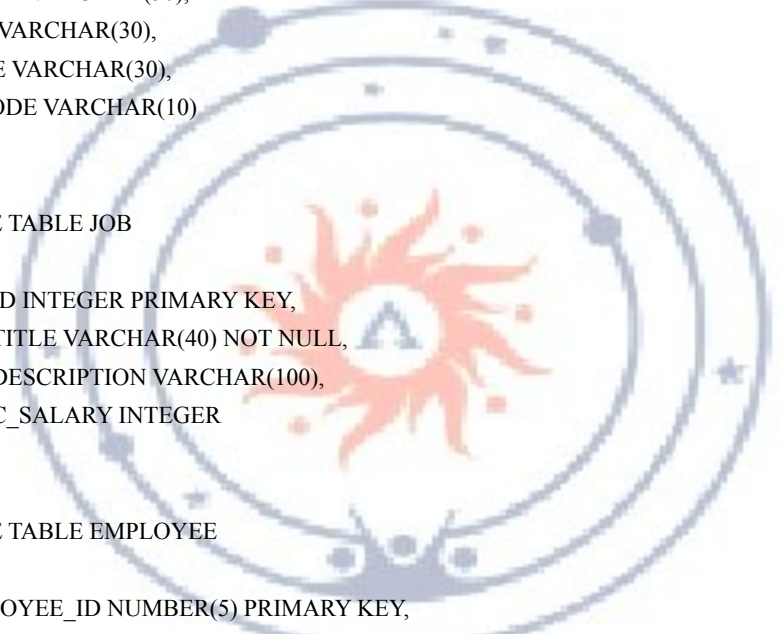
The following DDL is a documentation-ready implementation aligned with the supplied relational schema. Data types may be adjusted to the specific DBMS used by the student team.

```
CREATE TABLE DEPARTMENT
(
    DEPARTMENT_ID INTEGER PRIMARY KEY,
    DEPARTMENT_NAME VARCHAR(30) NOT NULL,
    STREET VARCHAR(50),
    CITY VARCHAR(30),
    STATE VARCHAR(30),
    ZIPCODE VARCHAR(10)
);

CREATE TABLE JOB
(
    JOB_ID INTEGER PRIMARY KEY,
    JOB_TITLE VARCHAR(40) NOT NULL,
    JOB_DESCRIPTION VARCHAR(100),
    BASIC_SALARY INTEGER
);

CREATE TABLE EMPLOYEE
(
    EMPLOYEE_ID NUMBER(5) PRIMARY KEY,
    FIRST_NAME VARCHAR(20) NOT NULL,
    MIDDLE_NAME VARCHAR(20) NOT NULL,
    LAST_NAME VARCHAR(20) NOT NULL,
    DOB DATE,
    EMAIL VARCHAR(50),
    DEPARTMENT_ID INTEGER,
    JOB_ID INTEGER,
    CONSTRAINT FK_EMP_DEPT FOREIGN KEY (DEPARTMENT_ID) REFERENCES
    DEPARTMENT(DEPARTMENT_ID),
    CONSTRAINT FK_EMP_JOB FOREIGN KEY (JOB_ID) REFERENCES JOB(JOB_ID)
);

CREATE TABLE EMPLOYEE_PHONE
(
```



```

    EMPLOYEE_ID INTEGER,
    PHONE VARCHAR(15),
    CONSTRAINT PK_EMP_PHONE PRIMARY KEY (EMPLOYEE_ID, PHONE),
    CONSTRAINT FK_PHONE_EMP FOREIGN KEY (EMPLOYEE_ID) REFERENCES
    EMPLOYEE(EMPLOYEE_ID)
);

```

```

CREATE TABLE ATTENDANCE
(
    ATTENDANCE_ID INTEGER PRIMARY KEY,
    EMPLOYEE_ID INTEGER NOT NULL,
    ATTENDANCE_DATE DATE NOT NULL,
    CHECKIN VARCHAR(10),
    CHECKOUT VARCHAR(10),
    STATUS VARCHAR(15),
    CONSTRAINT FK_ATT_EMP FOREIGN KEY (EMPLOYEE_ID) REFERENCES
    EMPLOYEE(EMPLOYEE_ID)
);

```

```

CREATE TABLE LEAVE
(
    LEAVE_ID INTEGER PRIMARY KEY,
    EMPLOYEE_ID INTEGER NOT NULL,
    LEAVE_TYPE VARCHAR(20),
    START_DATE DATE,
    END_DATE DATE,
    LEAVE_STATUS VARCHAR(20),
    REASON VARCHAR(100),
    CONSTRAINT FK_LEAVE_EMP FOREIGN KEY (EMPLOYEE_ID) REFERENCES
    EMPLOYEE(EMPLOYEE_ID)
);

```

```

CREATE TABLE PAYROLL
(
    PAYROLL_ID INTEGER PRIMARY KEY,
    EMPLOYEE_ID INTEGER NOT NULL,
    SALARY_MONTH VARCHAR(20),
    BASIC_SALARY INTEGER,
    BONUS INTEGER,
    DEDUCTION INTEGER,
    NET_SALARY INTEGER,
    CONSTRAINT FK_PAY_EMP FOREIGN KEY (EMPLOYEE_ID) REFERENCES
    EMPLOYEE(EMPLOYEE_ID)
);

```

## 8.2 DML – Data Manipulation Language

```
INSERT INTO DEPARTMENT VALUES
```

```
(101,'Human Resources' ,'MG Road','Hyderabad','Telangana','500001');
```

```
INSERT INTO EMPLOYEE
```

```
(Employee_ID, Employee_Name, Phone, Email, Department_ID, Job_ID)
```

```
VALUES(1001,'Ravi','Kumar','Reddy','98876500001','ravi.reddy@gmail.com',103,201);
```

```
INSERT INTO LEAVE
```

```
(Leave_ID, LeaveType, LeaveStatus, Reason, StartDate, EndDate, Employee_ID)
```

```
VALUES(401,'Casual',DATE '2026-08-05',DATE '2026-08-05','Approved','Personal work', 1001);
```

```
INSERT INTO PAYROLL
```

```
(Payroll_ID, SalaryMonth, BasicSalary, Bonus, Deduction, NetSalary, Employee_ID)
```

```
VALUES(501,'August 2026',60000,5000,2000,63000, 1001);
```

```
INSERT INTO JOB VALUES
```

```
(Job_ID, JobTitle, JobDescription, BasicSalary)
```

```
(201,'Software Engineer' ,'Develops software applications',60000);
```

## 8.3 DQL – Data Query Language

```
SELECT * FROM EMPLOYEE;
```

```
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL  
FROM EMPLOYEE
```

```
WHERE DEPARTMENT_ID = 103;
```

```
SELECT JOB_ID, JOB_TITLE, BASIC_SALARY
```

```
FROM JOB
```

```
ORDER BY JOB_TITLE;
```

```
SELECT E.FIRST_NAME, E.LAST_NAME, D.DEPARTMENT_NAME
```

```
FROM EMPLOYEE E
```

```
JOIN DEPARTMENT D
```

```
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

```
SELECT E.FIRST_NAME, E.LAST_NAME, J.JOB_TITLE, J.BASIC_SALARY
```

```
FROM EMPLOYEE E
```

```
JOIN JOB J
```

```
ON E.JOB_ID = J.JOB_ID;
```

## 9. SQL QUERIES & SCREENSHOTS

The following query set is prepared to satisfy the documentation requirement for a 30–40 query SQL section. The examples cover selection, filtering, joins, aggregation, subqueries, updates, deletion, and reporting. Exact outputs depend on the records inserted in the final project database.

### Q1. Display all departments

```
SELECT * FROM department;
```

**Output:**

| DEPARTMENT_ID | DEPARTMENT_NAME        | STREET          | CITY          | STATE          | ZIPCODE |
|---------------|------------------------|-----------------|---------------|----------------|---------|
| 102           | Finance                | Park Street     | Kolkata       | West Bengal    | 700016  |
| 103           | Information Technology | Banjara Road    | Hyderabad     | Telangana      | 500034  |
| 107           | Research               | Beach Road      | Visakhapatnam | Andhra Pradesh | 530001  |
| 108           | Customer Service       | Main Road       | Vijayawada    | Andhra Pradesh | 520001  |
| 111           | Production             | Industrial Road | Mumbai        | Maharashtra    | 400001  |
| 115           | Procurement            | Market Road     | Rajahmundry   | Andhra Pradesh | 533101  |
| 109           | Legal                  | Kasturba Road   | Bengaluru     | Karnataka      | 560001  |
| 110           | Administration         | Station Road    | Delhi         | Delhi          | 110001  |
| 114           | Security               | Airport Road    | Kochi         | Kerala         | 682001  |
| 101           | Human Resources        | MG Road         | Hyderabad     | Telangana      | 500001  |
| 104           | Marketing              | Anna Salai      | Chennai       | Tamil Nadu     | 600002  |
| 105           | Sales                  | MG Road         | Bengaluru     | Karnataka      | 560001  |
| 106           | Operations             | FC Road         | Pune          | Maharashtra    | 411004  |
| 112           | Quality Assurance      | Ring Road       | Ahmedabad     | Gujarat        | 380001  |
| 113           | Training               | College Road    | Coimbatore    | Tamil Nadu     | 641001  |

### Q2. Display all employee

```
SELECT * FROM Employee;
```

**Output:**

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                                                              | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|--------------------------------------------------------------------|---------------|--------|
| 1008        | Divya      | Priya       | Nair      | 2001-04-22T00:00:00Z | <a href="mailto:divya.nair@gmail.com">divya.nair@gmail.com</a>     | 107           | 208    |
| 1001        | Ravi       | Kumar       | Reddy     | 1999-05-12T00:00:00Z | <a href="mailto:ravi.reddy@gmail.com">ravi.reddy@gmail.com</a>     | 103           | 201    |
| 1002        | Priya      | Devi        | Sharma    | 2000-07-21T00:00:00Z | <a href="mailto:priya.sharma@gmail.com">priya.sharma@gmail.com</a> | 101           | 203    |
| 1004        | Sneha      | Lakshmi     | Patel     | 2001-11-10T00:00:00Z | <a href="mailto:sneha.patel@gmail.com">sneha.patel@gmail.com</a>   | 102           | 204    |
| 1005        | Vikram     | Raj         | Singh     | 1997-09-25T00:00:00Z | <a href="mailto:vikram.singh@gmail.com">vikram.singh@gmail.com</a> | 104           | 205    |
| 1006        | Anjali     | Sri         | Reddy     | 2000-01-18T00:00:00Z | <a href="mailto:anjali.reddy@gmail.com">anjali.reddy@gmail.com</a> | 105           | 206    |
| 1011        | Rahul      | Raj         | Verma     | 1998-10-19T00:00:00Z | <a href="mailto:rahul.verma@gmail.com">rahul.verma@gmail.com</a>   | 109           | 210    |
| 1012        | Pooja      | Rani        | Gupta     | 2000-02-27T00:00:00Z | <a href="mailto:pooja.gupta@gmail.com">pooja.gupta@gmail.com</a>   | 112           | 212    |
| 1014        | Swathi     | Sri         | Rao       | 2001-09-16T00:00:00Z | <a href="mailto:swathi.rao@gmail.com">swathi.rao@gmail.com</a>     | 114           | 213    |
| 1013        | Naveen     | Kumar       | Reddy     | 1997-01-09T00:00:00Z | <a href="mailto:naveen.reddy@gmail.com">naveen.reddy@gmail.com</a> | 113           | 211    |
| 1003        | Arjun      | Kumar       | Rao       | 1998-03-15T00:00:00Z | <a href="mailto:arjun.rao@gmail.com">arjun.rao@gmail.com</a>       | 103           | 202    |
| 1007        | Kiran      | Mohan       | Kumar     | 1996-06-30T00:00:00Z | <a href="mailto:kiran.kumar@gmail.com">kiran.kumar@gmail.com</a>   | 106           | 207    |
| 1009        | Suresh     | Babu        | Rao       | 1995-12-05T00:00:00Z | <a href="mailto:suresh.rao@gmail.com">suresh.rao@gmail.com</a>     | 103           | 201    |
| 1010        | Meena      | Kumari      | Das       | 1999-08-14T00:00:00Z | <a href="mailto:meena.das@gmail.com">meena.das@gmail.com</a>       | 108           | 214    |
| 1015        | Manoj      | Kumar       | Naidu     | 1996-05-28T00:00:00Z | <a href="mailto:manoj.naidu@gmail.com">manoj.naidu@gmail.com</a>   | 115           | 215    |

### Q3. Display all Jobs

```
SELECT * FROM Job;
```

Output:

| JOB_ID | JOB_TITLE                | JOB_DESCRIPTION                | BASIC_SALARY |
|--------|--------------------------|--------------------------------|--------------|
| 203    | HR Executive             | Manages employee activities    | 45000        |
| 208    | Data Analyst             | Analyzes business data         | 65000        |
| 210    | Business Analyst         | Analyzes business requirements | 70000        |
| 212    | Quality Analyst          | Tests software products        | 55000        |
| 201    | Software Engineer        | Develops software applications | 60000        |
| 202    | Senior Software Engineer | Handles advanced development   | 85000        |
| 209    | System Administrator     | Maintains IT systems           | 62000        |
| 206    | Sales Executive          | Handles sales operations       | 42000        |
| 207    | Project Manager          | Manages projects and teams     | 95000        |
| 214    | Support Executive        | Provides customer support      | 40000        |
| 205    | Marketing Executive      | Handles marketing activities   | 48000        |
| 213    | Database Administrator   | Manages databases              | 75000        |
| 204    | Accountant               | Manages financial records      | 50000        |
| 211    | Team Leader              | Leads development teams        | 80000        |
| 215    | Operations Manager       | Manages operations             | 90000        |

### Q4. Display all Payrolls

```
SELECT * FROM Payroll;
```

Output:

| PAYROLL_ID | EMPLOYEE_ID | SALARY_MONTH | BASIC_SALARY | BONUS | DEDUCTION | NET_SALARY |
|------------|-------------|--------------|--------------|-------|-----------|------------|
| 507        | 1007        | August 2026  | 95000        | 8000  | 4000      | 99000      |
| 508        | 1008        | August 2026  | 65000        | 5000  | 2000      | 68000      |
| 509        | 1009        | August 2026  | 62000        | 4500  | 1800      | 64700      |
| 510        | 1010        | August 2026  | 40000        | 2500  | 800       | 41700      |
| 511        | 1011        | August 2026  | 70000        | 6000  | 2500      | 73500      |
| 512        | 1012        | August 2026  | 55000        | 3500  | 1500      | 57000      |
| 513        | 1013        | August 2026  | 80000        | 6000  | 2500      | 83500      |
| 514        | 1014        | August 2026  | 75000        | 5000  | 2000      | 78000      |
| 515        | 1015        | August 2026  | 90000        | 7000  | 3500      | 93500      |
| 501        | 1001        | August 2026  | 60000        | 5000  | 2000      | 63000      |
| 502        | 1002        | August 2026  | 45000        | 3000  | 1000      | 47000      |
| 503        | 1003        | August 2026  | 85000        | 7000  | 3000      | 89000      |
| 504        | 1004        | August 2026  | 50000        | 4000  | 1500      | 52500      |
| 505        | 1005        | August 2026  | 48000        | 3500  | 1200      | 50300      |
| 506        | 1006        | August 2026  | 42000        | 2500  | 1000      | 43500      |

### Q5. Employees from a department

```
SELECT * FROM Employee WHERE Department_ID = 103;
```

Output:

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|----------------------|---------------|--------|
| 1001        | Ravi       | Kumar       | Reddy     | 1999-05-12T00:00:00Z | ravi.reddy@gmail.com | 103           | 201    |
| 1003        | Arjun      | Kumar       | Rao       | 1998-03-15T00:00:00Z | arjun.rao@gmail.com  | 103           | 202    |
| 1009        | Suresh     | Babu        | Rao       | 1995-12-05T00:00:00Z | suresh.rao@gmail.com | 103           | 201    |

### Q6. Jobs with Basic Salary greater than 60000

```
SELECT * FROM Job WHERE Basic Salary > 60000;
```

Output:

| JOB_ID | JOB_TITLE                | JOB_DESCRIPTION                | BASIC_SALARY |
|--------|--------------------------|--------------------------------|--------------|
| 208    | Data Analyst             | Analyzes business data         | 65000        |
| 210    | Business Analyst         | Analyzes business requirements | 70000        |
| 202    | Senior Software Engineer | Handles advanced development   | 85000        |
| 209    | System Administrator     | Maintains IT systems           | 62000        |
| 207    | Project Manager          | Manages projects and teams     | 95000        |
| 213    | Database Administrator   | Manages databases              | 75000        |
| 211    | Team Leader              | Leads development teams        | 80000        |
| 215    | Operations Manager       | Manages operations             | 90000        |

### Q7. Employees sorted by first name

```
SELECT * FROM Employee ORDER BY First_Name;
```

Output:

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                  | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|------------------------|---------------|--------|
| 1006        | Anjali     | Sri         | Reddy     | 2000-01-18T00:00:00Z | anjali.reddy@gmail.com | 105           | 206    |
| 1003        | Arjun      | Kumar       | Rao       | 1998-03-15T00:00:00Z | arjun.rao@gmail.com    | 103           | 202    |
| 1008        | Divya      | Priya       | Nair      | 2001-04-22T00:00:00Z | divya.nair@gmail.com   | 107           | 208    |
| 1007        | Kiran      | Mohan       | Kumar     | 1996-06-30T00:00:00Z | kiran.kumar@gmail.com  | 106           | 207    |
| 1015        | Manoj      | Kumar       | Naidu     | 1996-05-28T00:00:00Z | manoj.naidu@gmail.com  | 115           | 215    |
| 1010        | Meena      | Kumari      | Das       | 1999-08-14T00:00:00Z | meena.das@gmail.com    | 108           | 214    |
| 1013        | Naveen     | Kumar       | Reddy     | 1997-01-09T00:00:00Z | naveen.reddy@gmail.com | 113           | 211    |
| 1012        | Pooja      | Rani        | Gupta     | 2000-02-27T00:00:00Z | pooja.gupta@gmail.com  | 112           | 212    |
| 1002        | Priya      | Devi        | Sharma    | 2000-07-21T00:00:00Z | priya.sharma@gmail.com | 101           | 203    |
| 1011        | Rahul      | Raj         | Verma     | 1998-10-19T00:00:00Z | rahul.verma@gmail.com  | 109           | 210    |
| 1001        | Ravi       | Kumar       | Reddy     | 1999-05-12T00:00:00Z | ravi.reddy@gmail.com   | 103           | 201    |
| 1004        | Sneha      | Lakshmi     | Patel     | 2001-11-10T00:00:00Z | sneha.patel@gmail.com  | 102           | 204    |
| 1009        | Suresh     | Babu        | Rao       | 1995-12-05T00:00:00Z | suresh.rao@gmail.com   | 103           | 201    |
| 1014        | Swathi     | Sri         | Rao       | 2001-09-16T00:00:00Z | swathi.rao@gmail.com   | 114           | 213    |
| 1005        | Vikram     | Raj         | Singh     | 1997-09-25T00:00:00Z | vikram.singh@gmail.com | 104           | 205    |

### Q8. Jobs sorted by Basic Salary

```
SELECT * FROM Job ORDER BY Basic_Salary DESC;
```

Output:

| JOB_ID | JOB_TITLE                | JOB_DESCRIPTION                | BASIC_SALARY |
|--------|--------------------------|--------------------------------|--------------|
| 207    | Project Manager          | Manages projects and teams     | 95000        |
| 215    | Operations Manager       | Manages operations             | 90000        |
| 202    | Senior Software Engineer | Handles advanced development   | 85000        |
| 211    | Team Leader              | Leads development teams        | 80000        |
| 213    | Database Administrator   | Manages databases              | 75000        |
| 210    | Business Analyst         | Analyzes business requirements | 70000        |
| 208    | Data Analyst             | Analyzes business data         | 65000        |
| 209    | System Administrator     | Maintains IT systems           | 62000        |
| 201    | Software Engineer        | Develops software applications | 60000        |
| 212    | Quality Analyst          | Tests software products        | 55000        |
| 204    | Accountant               | Manages financial records      | 50000        |
| 205    | Marketing Executive      | Handles marketing activities   | 48000        |
| 203    | HR Executive             | Manages employee activities    | 45000        |
| 206    | Sales Executive          | Handles sales operations       | 42000        |
| 214    | Support Executive        | Provides customer support      | 40000        |

### Q9. Employees with email addresses

```
SELECT Employee_ID, First_Name, Last_Name, Email FROM Employee WHERE Email IS NOT NULL;
```

Output:

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL                                                              |
|-------------|------------|-----------|--------------------------------------------------------------------|
| 1008        | Divya      | Nair      | <a href="mailto:divya.nair@gmail.com">divya.nair@gmail.com</a>     |
| 1001        | Ravi       | Reddy     | <a href="mailto:ravi.reddy@gmail.com">ravi.reddy@gmail.com</a>     |
| 1002        | Priya      | Sharma    | <a href="mailto:priya.sharma@gmail.com">priya.sharma@gmail.com</a> |
| 1004        | Sneha      | Patel     | <a href="mailto:sneha.patel@gmail.com">sneha.patel@gmail.com</a>   |
| 1005        | Vikram     | Singh     | <a href="mailto:vikram.singh@gmail.com">vikram.singh@gmail.com</a> |
| 1006        | Anjali     | Reddy     | <a href="mailto:anjali.reddy@gmail.com">anjali.reddy@gmail.com</a> |
| 1011        | Rahul      | Verma     | <a href="mailto:rahul.verma@gmail.com">rahul.verma@gmail.com</a>   |
| 1012        | Pooja      | Gupta     | <a href="mailto:pooja.gupta@gmail.com">pooja.gupta@gmail.com</a>   |
| 1014        | Swathi     | Rao       | <a href="mailto:swathi.rao@gmail.com">swathi.rao@gmail.com</a>     |
| 1013        | Naveen     | Reddy     | <a href="mailto:naveen.reddy@gmail.com">naveen.reddy@gmail.com</a> |
| 1003        | Arjun      | Rao       | <a href="mailto:arjun.rao@gmail.com">arjun.rao@gmail.com</a>       |
| 1007        | Kiran      | Kumar     | <a href="mailto:kiran.kumar@gmail.com">kiran.kumar@gmail.com</a>   |
| 1009        | Suresh     | Rao       | <a href="mailto:suresh.rao@gmail.com">suresh.rao@gmail.com</a>     |
| 1010        | Meena      | Das       | <a href="mailto:meena.das@gmail.com">meena.das@gmail.com</a>       |
| 1015        | Manoj      | Naidu     | <a href="mailto:manoj.naidu@gmail.com">manoj.naidu@gmail.com</a>   |

### Q10. Department and Employee

```
SELECT D.Department_Name, E.Firstt_Name, E.Last_Name  
FROM Department D JOIN Employee E ON D.Department_ID = E.Department_ID;
```

Output:

| DEPARTMENT_NAME        | FIRST_NAME | LAST_NAME |
|------------------------|------------|-----------|
| Research               | Divya      | Nair      |
| Information Technology | Ravi       | Reddy     |
| Human Resources        | Priya      | Sharma    |
| Finance                | Sneha      | Patel     |
| Marketing              | Vikram     | Singh     |
| Sales                  | Anjali     | Reddy     |
| Legal                  | Rahul      | Verma     |
| Quality Assurance      | Pooja      | Gupta     |
| Security               | Swathi     | Rao       |
| Training               | Naveen     | Reddy     |
| Information Technology | Arjun      | Rao       |
| Operations             | Kiran      | Kumar     |
| Information Technology | Suresh     | Rao       |
| Customer Service       | Meena      | Das       |
| Procurement            | Manoj      | Naidu     |

### Q11. Employees with department

```
SELECT E.First_Name,E.Last_Name, D.Department_Name  
FROM Employee E JOIN Department D ON E.Department_ID = D.Department_ID;
```

Output:

| FIRST_NAME | LAST_NAME | DEPARTMENT_NAME        |
|------------|-----------|------------------------|
| Sneha      | Patel     | Finance                |
| Ravi       | Reddy     | Information Technology |
| Arjun      | Rao       | Information Technology |
| Suresh     | Rao       | Information Technology |
| Divya      | Nair      | Research               |
| Meena      | Das       | Customer Service       |
| Manoj      | Naidu     | Procurement            |
| Rahul      | Verma     | Legal                  |
| Swathi     | Rao       | Security               |
| Priya      | Sharma    | Human Resources        |
| Vikram     | Singh     | Marketing              |
| Anjali     | Reddy     | Sales                  |
| Kiran      | Kumar     | Operations             |
| Pooja      | Gupta     | Quality Assurance      |
| Naveen     | Reddy     | Training               |

## Q12. Employees with Job

```
SELECT E.First_Name, E.Last_Name, J.Job_Title
FROM Employee E JOIN Job J ON E.Job_ID = J.Job_ID;
```

**Output:**

| FIRST_NAME | LAST_NAME | JOB_TITLE                |
|------------|-----------|--------------------------|
| Priya      | Sharma    | HR Executive             |
| Divya      | Nair      | Data Analyst             |
| Rahul      | Verma     | Business Analyst         |
| Pooja      | Gupta     | Quality Analyst          |
| Ravi       | Reddy     | Software Engineer        |
| Suresh     | Rao       | Software Engineer        |
| Arjun      | Rao       | Senior Software Engineer |
| Anjali     | Reddy     | Sales Executive          |
| Kiran      | Kumar     | Project Manager          |
| Meena      | Das       | Support Executive        |
| Vikram     | Singh     | Marketing Executive      |
| Swathi     | Rao       | Database Administrator   |
| Sneha      | Patel     | Accountant               |
| Naveen     | Reddy     | Team Leader              |
| Manoj      | Naidu     | Operations Manager       |

## Q13. Employee with Payroll details

```
SELECT E.First_Name, E.Last_Name, P.Basic_Salary, P.Bonus, P.Deduction,
P.Net_Salary
FROM Employee E JOIN Payroll P ON E.Employee_ID = P.Employee_ID;
```

**Output:**

| FIRST_NAME | LAST_NAME | BASIC_SALARY | BONUS | DEDUCTION | NET_SALARY |
|------------|-----------|--------------|-------|-----------|------------|
| Kiran      | Kumar     | 95000        | 8000  | 4000      | 99000      |
| Divya      | Nair      | 65000        | 5000  | 2000      | 68000      |
| Suresh     | Rao       | 62000        | 4500  | 1800      | 64700      |
| Meena      | Das       | 40000        | 2500  | 800       | 41700      |
| Rahul      | Verma     | 70000        | 6000  | 2500      | 73500      |
| Pooja      | Gupta     | 55000        | 3500  | 1500      | 57000      |
| Naveen     | Reddy     | 80000        | 6000  | 2500      | 83500      |
| Swathi     | Rao       | 75000        | 5000  | 2000      | 78000      |
| Manoj      | Naidu     | 90000        | 7000  | 3500      | 93500      |
| Ravi       | Reddy     | 60000        | 5000  | 2000      | 63000      |
| Priya      | Sharma    | 45000        | 3000  | 1000      | 47000      |
| Arjun      | Rao       | 85000        | 7000  | 3000      | 89000      |
| Sneha      | Patel     | 50000        | 4000  | 1500      | 52500      |
| Vikram     | Singh     | 48000        | 3500  | 1200      | 50300      |
| Anjali     | Reddy     | 42000        | 2500  | 1000      | 43500      |

#### Q14. Employee Attendance Details

```
SELECT E.First_Name, E.Last_Name, A.Attendance_Data, A.Status from Employee
E
Join Attendance A on E.Employee_ID =A.Employee_ID;
```

**Output:**

| FIRST_NAME | LAST_NAME | ATTENDANCE_DATE      | STATUS  |
|------------|-----------|----------------------|---------|
| Ravi       | Reddy     | 2026-09-01T00:00:00Z | Present |
| Priya      | Sharma    | 2026-09-01T00:00:00Z | Present |
| Arjun      | Rao       | 2026-09-01T00:00:00Z | Present |
| Sneha      | Patel     | 2026-09-01T00:00:00Z | Late    |
| Vikram     | Singh     | 2026-09-01T00:00:00Z | Present |
| Anjali     | Reddy     | 2026-09-01T00:00:00Z | Present |
| Kiran      | Kumar     | 2026-09-01T00:00:00Z | Late    |
| Divya      | Nair      | 2026-09-01T00:00:00Z | Present |
| Suresh     | Rao       | 2026-09-01T00:00:00Z | Present |
| Meena      | Das       | 2026-09-01T00:00:00Z | Present |
| Rahul      | Verma     | 2026-09-01T00:00:00Z | Late    |
| Pooja      | Gupta     | 2026-09-01T00:00:00Z | Present |
| Naveen     | Reddy     | 2026-09-01T00:00:00Z | Present |
| Swathi     | Rao       | 2026-09-01T00:00:00Z | Late    |
| Manoj      | Naidu     | 2026-09-01T00:00:00Z | Present |

#### Q15. Employee Leave details

```
SELECT E.First_Name, E.Last_Name,L.Leave_Type, L.Leave_Status, L.Start_Date,
L.End_Date FROM Employee E JOIN Leave L ON E.Employee_ID = L.Employee_ID;
```

**Output:**

| FIRST_NAME | LAST_NAME | LEAVE_TYPE | LEAVE_STATUS | START_DATE           | END_DATE             |
|------------|-----------|------------|--------------|----------------------|----------------------|
| Priya      | Sharma    | Sick       | Approved     | 2026-08-08T00:00:00Z | 2026-08-09T00:00:00Z |
| Arjun      | Rao       | Casual     | Approved     | 2026-08-12T00:00:00Z | 2026-08-12T00:00:00Z |
| Sneha      | Patel     | Sick       | Pending      | 2026-08-15T00:00:00Z | 2026-08-15T00:00:00Z |
| Vikram     | Singh     | Casual     | Approved     | 2026-08-18T00:00:00Z | 2026-08-19T00:00:00Z |
| Anjali     | Reddy     | Earned     | Approved     | 2026-08-20T00:00:00Z | 2026-08-22T00:00:00Z |
| Kiran      | Kumar     | Casual     | Rejected     | 2026-08-21T00:00:00Z | 2026-08-21T00:00:00Z |
| Divya      | Nair      | Sick       | Approved     | 2026-08-23T00:00:00Z | 2026-08-24T00:00:00Z |
| Suresh     | Rao       | Earned     | Approved     | 2026-08-25T00:00:00Z | 2026-08-26T00:00:00Z |
| Meena      | Das       | Casual     | Approved     | 2026-08-27T00:00:00Z | 2026-08-27T00:00:00Z |
| Rahul      | Verma     | Sick       | Pending      | 2026-08-28T00:00:00Z | 2026-08-28T00:00:00Z |
| Pooja      | Gupta     | Earned     | Approved     | 2026-08-29T00:00:00Z | 2026-08-30T00:00:00Z |
| Naveen     | Reddy     | Casual     | Approved     | 2026-08-30T00:00:00Z | 2026-08-30T00:00:00Z |
| Swathi     | Rao       | Sick       | Approved     | 2026-08-31T00:00:00Z | 2026-08-31T00:00:00Z |
| Manoj      | Naidu     | Earned     | Pending      | 2026-09-01T00:00:00Z | 2026-09-02T00:00:00Z |
| Ravi       | Reddy     | Casual     | Approved     | 2026-08-05T00:00:00Z | 2026-08-05T00:00:00Z |

### Q16. Employees per department

```
SELECT Department_ID, COUNT(*) AS Employee_Count FROM Employee GROUP BY  
Department_ID;
```

**Output:**

| DEPARTMENT_ID | EMPLOYEE_COUNT |
|---------------|----------------|
| 107           | 1              |
| 103           | 3              |
| 101           | 1              |
| 102           | 1              |
| 104           | 1              |
| 105           | 1              |
| 109           | 1              |
| 112           | 1              |
| 114           | 1              |
| 113           | 1              |
| 106           | 1              |
| 108           | 1              |
| 115           | 1              |

### Q17. Employees per Job

```
SELECT Job_ID, COUNT(*) AS Employee_Count FROM Employee GROUP BY Job_ID;
```

**Output:**

| JOB_ID | EMPLOYEE_COUNT |
|--------|----------------|
| 208    | 1              |
| 201    | 2              |
| 203    | 1              |
| 204    | 1              |
| 205    | 1              |
| 206    | 1              |
| 210    | 1              |
| 212    | 1              |
| 213    | 1              |
| 211    | 1              |
| 202    | 1              |
| 207    | 1              |
| 214    | 1              |
| 215    | 1              |

### Q18. Average Net Salary

```
SELECT AVG(Net_Salary) AS Average_Salary FROM Payroll;
```

**Output:**

| AVERAGE_SALARY    |
|-------------------|
| 66946.66666666670 |

### Q19. Maximum Net Salary

```
SELECT MAX(Net_Salary) AS Highest_Marks FROM Payroll;
```

**Output:**

| HIGHEST_SALARY |
|----------------|
| 99000          |

### Q20. Minimum Net Salary

```
SELECT MIN(Net_Salary) AS Lowest_Marks FROM Payroll;
```

**Output:**

| LOWEST_MARKS |
|--------------|
| 41700        |

### Q21. Total Salary Paid

```
SELECT SUM(Net_Salary) AS Total_Salary FROM Payroll;
```

**Output:**

| TOTAL_SALARY |
|--------------|
| 1004200      |

### Q22. Count of Employees

```
SELECT COUNT(*) AS Total_Employees FROM Employee;
```

**Output:**

| TOTAL_EMPLOYEES |
|-----------------|
| 15              |

### Q23. Employees with Pending Leave

```
SELECT E.First_Name, E.Last_Name FROM Employee E Join Leave L  
On E.Employee_ID = L.Employee_ID WHERE L.Leave_Status='Pending';
```

**Output:**

| FIRST_NAME | LAST_NAME |
|------------|-----------|
| Sneha      | Patel     |
| Rahul      | Verma     |
| Manoj      | Naidu     |

### Q24. Employees with Late attendance

```
SELECT E.First_Name, E.Last_Name, A.Attendance_Date FROM Employee E  
JOIN Attendance A ON E.Employee_ID = A.Employee_ID WHERE A.STATUS = 'Late';
```

**Output:**

| FIRST_NAME | LAST_NAME | ATTENDANCE_DATE      |
|------------|-----------|----------------------|
| Sneha      | Patel     | 2026-09-01T00:00:00Z |
| Rahul      | Verma     | 2026-09-01T00:00:00Z |
| Swathi     | Rao       | 2026-09-01T00:00:00Z |
| Kiran      | Kumar     | 2026-09-01T00:00:00Z |

### Q25. Employees with salary above 70000

```
SELECT E.First_Name, E.Last_Name, P.Net_Salary From Employee E  
Join Payroll P ON E.Employee_ID = P.Employee_ID where P.Net_Salary > 70000 ;
```

**Output:**

| FIRST_NAME | LAST_NAME | NET_SALARY |
|------------|-----------|------------|
| Rahul      | Verma     | 73500      |
| Swathi     | Rao       | 78000      |
| Naveen     | Reddy     | 83500      |
| Arjun      | Rao       | 89000      |
| Kiran      | Kumar     | 99000      |
| Manoj      | Naidu     | 93500      |

### Q26. Department with their Employee Count

```
SELECT D.DEPARTMENT_NAME, COUNT(E.EMPLOYEE_ID) AS EMPLOYEE_COUNT FROM DEPARTMENT  
D  
  
JOIN EMPLOYEE E ON D.DEPARTMENT_ID = E.DEPARTMENT_ID GROUP BY  
D.DEPARTMENT_NAME;
```

**Output:**

| DEPARTMENT_NAME        | EMPLOYEE_COUNT |
|------------------------|----------------|
| Research               | 1              |
| Information Technology | 3              |
| Human Resources        | 1              |
| Finance                | 1              |
| Marketing              | 1              |
| Sales                  | 1              |
| Legal                  | 1              |
| Quality Assurance      | 1              |
| Security               | 1              |
| Training               | 1              |
| Operations             | 1              |
| Customer Service       | 1              |
| Procurement            | 1              |

### Q27. Jobs with Basic Salary above 60000

```
SELECT Job_Title, Basic_Salary FROM Job WHERE Basic_Salary > 60000;
```

**Output:**

| JOB_TITLE                | BASIC_SALARY |
|--------------------------|--------------|
| Data Analyst             | 65000        |
| Business Analyst         | 70000        |
| Senior Software Engineer | 85000        |
| System Administrator     | 62000        |
| Project Manager          | 95000        |
| Database Administrator   | 75000        |
| Team Leader              | 80000        |
| Operations Manager       | 90000        |

### Q28. Employees whose names start with A

```
SELECT * FROM Employee WHERE First_Name LIKE 'A%';
```

**Output:**

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                                                              | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|--------------------------------------------------------------------|---------------|--------|
| 1006        | Anjali     | Sri         | Reddy     | 2000-01-18T00:00:00Z | <a href="mailto:anjali.reddy@gmail.com">anjali.reddy@gmail.com</a> | 105           | 206    |
| 1003        | Arjun      | Kumar       | Rao       | 1998-03-15T00:00:00Z | <a href="mailto:arjun.rao@gmail.com">arjun.rao@gmail.com</a>       | 103           | 202    |

### Q29. Search Employees by name

```
SELECT * FROM Employee WHERE UPPER(First_Name) LIKE '%RA%';
```

**Output:**

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                                                              | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|--------------------------------------------------------------------|---------------|--------|
| 1001        | Ravi       | Kumar       | Reddy     | 1999-05-12T00:00:00Z | <a href="mailto:ravi.reddy@gmail.com">ravi.reddy@gmail.com</a>     | 103           | 201    |
| 1005        | Vikram     | Raj         | Singh     | 1997-09-25T00:00:00Z | <a href="mailto:vikram.singh@gmail.com">vikram.singh@gmail.com</a> | 104           | 205    |
| 1011        | Rahul      | Raj         | Verma     | 1998-10-19T00:00:00Z | <a href="mailto:rahul.verma@gmail.com">rahul.verma@gmail.com</a>   | 109           | 210    |
| 1007        | Kiran      | Mohan       | Kumar     | 1996-06-30T00:00:00Z | <a href="mailto:kiran.kumar@gmail.com">kiran.kumar@gmail.com</a>   | 106           | 207    |

### Q30. Employees without department

```
SELECT * FROM Employee WHERE Department_ID IS NULL;
```

**Output:**

| EMPLOYEE_ID | FIRST_NAME | MIDDLE_NAME | LAST_NAME | DOB                  | EMAIL                                                              | DEPARTMENT_ID | JOB_ID |
|-------------|------------|-------------|-----------|----------------------|--------------------------------------------------------------------|---------------|--------|
| 1001        | Ravi       | Kumar       | Reddy     | 1999-05-12T00:00:00Z | <a href="mailto:ravi.reddy@gmail.com">ravi.reddy@gmail.com</a>     | 103           | 201    |
| 1005        | Vikram     | Raj         | Singh     | 1997-09-25T00:00:00Z | <a href="mailto:vikram.singh@gmail.com">vikram.singh@gmail.com</a> | 104           | 205    |
| 1011        | Rahul      | Raj         | Verma     | 1998-10-19T00:00:00Z | <a href="mailto:rahul.verma@gmail.com">rahul.verma@gmail.com</a>   | 109           | 210    |
| 1007        | Kiran      | Mohan       | Kumar     | 1996-06-30T00:00:00Z | <a href="mailto:kiran.kumar@gmail.com">kiran.kumar@gmail.com</a>   | 106           | 207    |



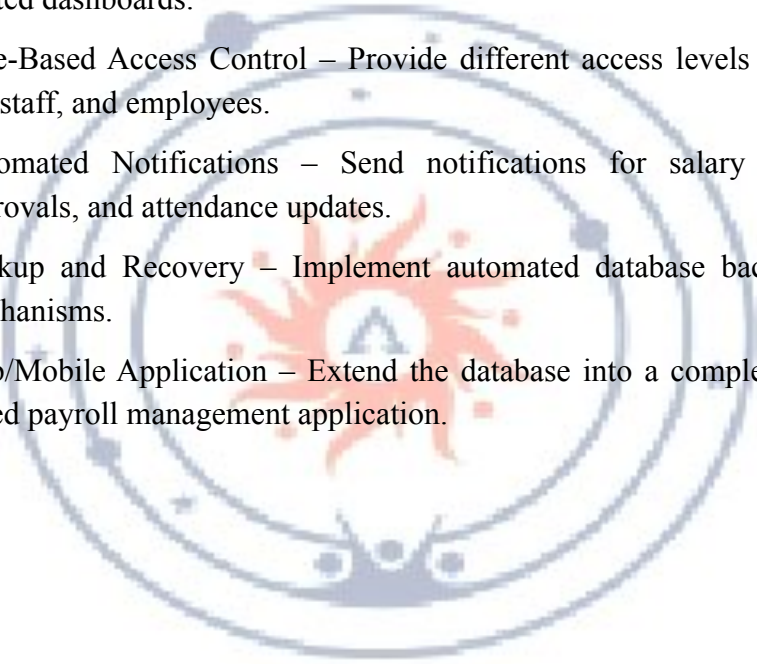
## 11. CONCLUSION

The Employee Payroll Management System provides a structured relational database solution for managing employee-related information. The design organizes departments, employees, jobs, attendance, leave, and payroll into related tables and uses primary keys, foreign keys, NOT NULL, UNIQUE, and CHECK constraints to maintain data integrity. The normalization process up to 3NF helps reduce data redundancy and update anomalies. The SQL implementation and query set demonstrate data insertion, modification, retrieval, aggregation, joins, and payroll-related reporting. The project therefore provides a suitable DBMS implementation that can be extended into a complete employee management and payroll system.



## 12. FUTURE ENHANCEMENTS

- Employee Self-Service Portal – Allow employees to view payroll, attendance, and leave details.
- Automated Attendance Management – Integrate biometric or digital attendance systems.
- Leave Management Enhancement – Provide online leave application, approval, and notification features.
- Payroll Reports and Analytics – Generate monthly salary reports and salary-related dashboards.
- Role-Based Access Control – Provide different access levels for administrators, HR staff, and employees.
- Automated Notifications – Send notifications for salary processing, leave approvals, and attendance updates.
- Backup and Recovery – Implement automated database backup and recovery mechanisms.
- Web/Mobile Application – Extend the database into a complete web or mobile-based payroll management application.



## REFERENCES

1. Korth, H. F., Silberschatz, A., and Sudarshan, S., Database System Concepts, McGraw Hill.
2. Elmasri, R. and Navathe, S. B., Fundamentals of Database Systems, Pearson.
3. Oracle Corporation, Oracle Database SQL Language Reference.
4. W3Schools, SQL Tutorial, SQL concepts, commands, queries, and database examples.
5. GeeksforGeeks, DBMS and SQL Tutorials, database concepts, SQL queries, keys, constraints, and normalization.
6. TutorialsPoint, DBMS and SQL Tutorial, database management concepts and SQL operations.
7. Course notes and laboratory materials provided for the Database Management Systems (DBMS) course.

